Chapter: 3

Time domain analysis of Control System

Transient Response and Steady-State Response: The time response of a control system consists of two parts: the transient response and the steady-state response. By transient response, we mean that which goes from the initial state to the final state. By steady-state response, we mean 'the manner in which the system output behaves as t approaches infinity. Thus the system response c (t) may be written as

$$C(t) = c_{tr}(t) + c_{ss}(t)$$

time domain / frequency domain output = transient response + s.s. response

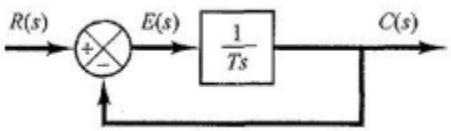
FIRST -ORDER SYSTEMS

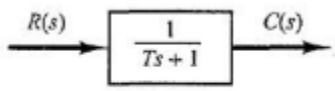
Consider the first-order system shown in Figure. The input-output relationship is

$$\frac{C(s)}{R(s)} = \frac{1}{Ts+1} \qquad (1)$$

order = the highest order of denominator
ex. 1/(s+1), 1/(s^2 + as +1), (s+1)/(s+1)(s+2)

Note that all systems having the same transfer function will exhibit the same output in response to the same input. For any given physical system, the mathematical response can be given a physical interpretation.





Unit-Step Response of First-Order Systems:

Since the Laplace transform of the unit-step function is l/s, substituting R(s) = l/s into Equation (1), we obtain

$$C(s) = \frac{1}{Ts+1}\frac{1}{s} = \frac{1}{s} - \frac{T}{Ts+1} = \frac{1}{s} - \frac{1}{s+(1/T)} \qquad (2)$$

T = Time constant ,
C(T) = (1-1/e) R(T),

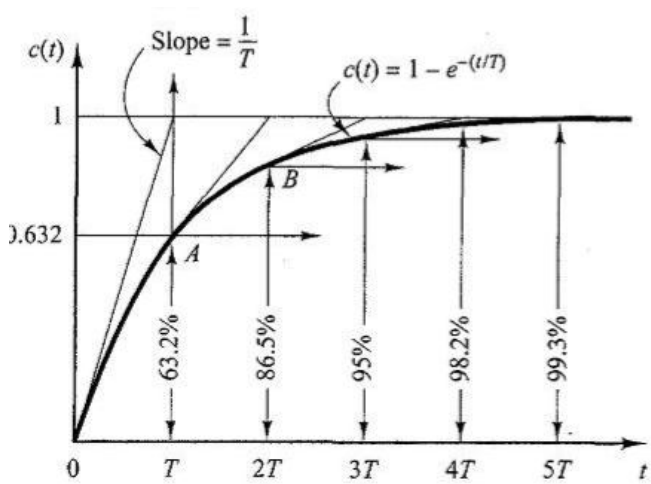
Taking the inverse Laplace transform of Equation (2), we obtain

$$c(t) = 1 - e^{-t/T}, \text{ for } t \geq 0 \qquad (3)$$

Equation (3) states that initially the output c(t) is zero and finally it becomes unity. One important characteristic of such an exponential response curve c(t) is that at t =T the value of c(t) is 0.632, or the response c(t) has reached 63.2% of its total change. This may be easily seen by substituting t = T in c (t). That is,

$$c(T) = 1 - e^{-1} = 0.632$$

standard test input



For $t \geq 4T$, the response remains within 2% of the final value. As seen from Equation (3), the steady- state is reached mathematically only after an infinite time. In practice, however, a reasonable estimate of the response time is the length of time the response curve needs to reach and stay within the 2% line of the final value, or four time constants.

Unit-Ramp Response of First-Order Systems:

Since the Laplace transform of the unit-ramp function is $\frac{1}{s^2}$,

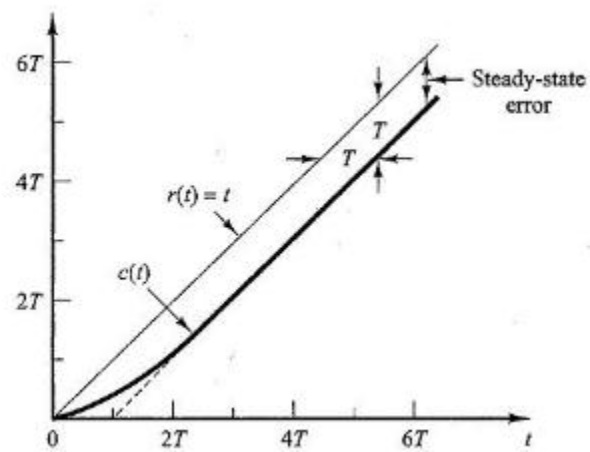
$$C(s) = \frac{1}{Ts+1}\frac{1}{s^2} = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts+1} \qquad (5)$$

Taking the inverse Laplace transform of Equation (5), we obtain

$$c(t) = t - T - Te^{-\frac{t}{T}}, \quad for \;\; t \geq 0 \qquad (6)$$

The error signal $e(t)$ is then

$$e(t) = r(t) - c(t)$$
$$= T(1 - e^{-t/T})$$



As t approaches infinity, $e^{-t/T}$ approaches zero, and thus the error signal e (t) approaches T or e (∞) = T, The smaller the time constant T, the smaller the steady-state error in the ramp input.

SECOND-ORDER SYSTEMS:

The standard form of the second-order system.

$$\frac{C(s)}{R(s)} = \frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+ {\omega_n}^2}$$

where ω_n, the undamped natural frequency; and ζ, the damping ratio of the system.

The dynamic behavior of the second-order system can then be described in terms of two parameters ζ andω_n.

If 0 < ζ < 1, the closed-loop poles are complex conjugates and lie in the left-half s plane. The system is then called underdamped, and the transient response is oscillatory. If ζ = 0, the transient response does not die out. If ζ = 1, the system is called critically damped. Overdamped systems correspond to ζ > 1.

Step Response of Second-Order System

(1). **Underdampedcase (0 < ζ < 1)**: In this case, C(s)/R(s) can be written

$$\frac{C(s)}{R(s)} = \frac{{\omega_n}^2}{(s + \zeta\omega_n + j\omega_d)(s + \zeta\omega_n - j\omega_d)} \qquad (A.1)$$

where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ The frequency ω_d is called the damped natural frequency. For a unit-step input, (C(s) can be written

$$C(s) = \frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+ {\omega_n}^2}\frac{1}{s}$$

The inverse Laplace transform can be obtained easily if C(s) is written in the form:

$$C(s) = \frac{1}{s} - \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+ {\omega_n}^2} = \frac{1}{s} - \frac{s+\zeta\omega_n}{(s+\zeta\omega_n)^2+{\omega_d}^2} - \frac{\zeta\omega_n}{(s+\zeta\omega_n)^2+{\omega_d}^2} \qquad (A.2)$$

i) Step Input (Position function) :

It is the sudden application of the input at a specified time as shown in the Fig. 7.2.

Mathematically it can be described as,

$$\boxed{\begin{aligned} r(t) &= A && \text{for } t \geq 0 \\ &= 0 && \text{for } t < 0 \end{aligned}}$$

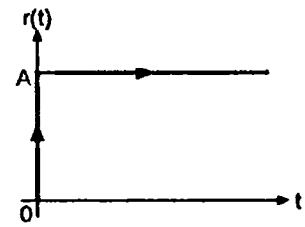


Fig. 7.2 Step

If A = 1, then it is called **unit step function** and denoted by u(t).

Laplace transform of such input is $\frac{A}{s}$.

ii) Ramp Input (Velocity function) :

It is constant rate of change in input i.e. gradual application of input as shown in the Fig. 7.3.

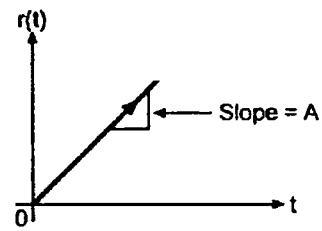
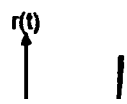


Fig. 7.3 Ramp

Magnitude of Ramp input is nothing but its slope. Mathematically it is defined as,

$$\boxed{\begin{aligned} r(t) &= At && \text{for } t \geq 0 \\ &= 0 && \text{for } t < 0 \end{aligned}}$$

If A = 1, it is called **Unit Ramp input**. It is denoted as r(t). Its Laplace transform is $\frac{A}{s^2}$.



$$\mathscr{L}^{-1}[C(s)] = c(t) = 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right) \qquad \text{(A.3)}$$

The error signal for this system is the difference between the input and output and is

$$e(t) = r(t) - c(t) = e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right), for\ t \geq 0$$

This error signal exhibits a damped sinusoidal oscillation. At steady state, or at t =∞, no error exists between the input and output. If the damping ratio ζ is equal to zero, the response becomes undamped and oscillations continue indefinitely.

$$e(t) = 1 - \cos\omega_n t\,, \quad for\ t \geq 0$$

(2). **Critically damped case (ζ = 1):** If the two poles of C(s)/R(s) are equal, the system is said to be a critically damped one. For a unit-step input,

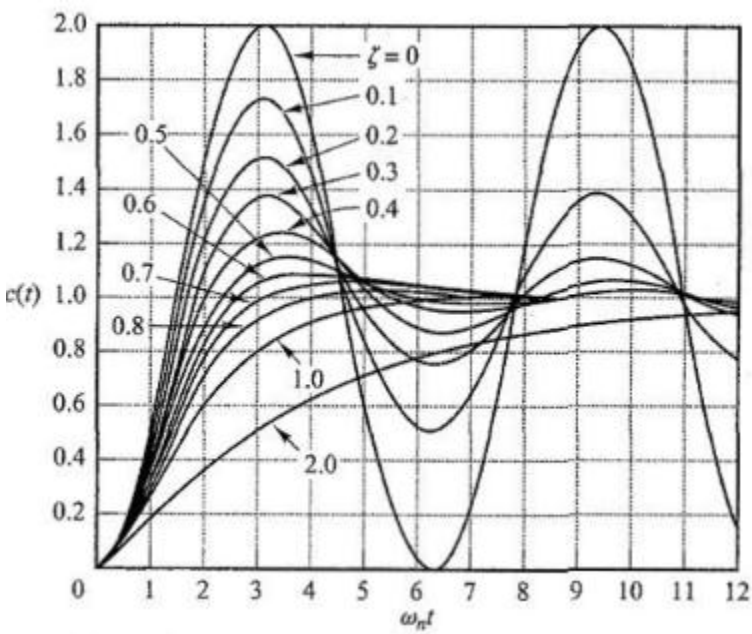
$$(C) = \frac{{\omega_n}^2}{(s+\omega_n)^2 s} \qquad c(t) = 1 - e^{-\omega_n t}(1+\omega_n t), \qquad for\ t \geq 0 \qquad \text{(A.4)}$$

(3). **Overdamped case (ζ > 1):** In this case, the two poles of C(s)/R(s) are negative real and unequal. For a unit-step input, R(s) = l/s and C(s) can be written

$$C(s) = \frac{{\omega_n}^2}{(s+\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})(s+\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})s}$$

An approximate unit-step response when one of the poles of C(s)/R(s) can be neglected.

$$c(t) = 1 - e^{-\left(\zeta-\sqrt{\zeta^2-1}\right)\omega_n t}, \quad for\ t \geq 0$$



Definitions of Transient-Response Specifications

The transient response of a practical control system often exhibits damped oscillations before reaching steady state. In specifying the transient-response characteristics of a control system to a unit-step input, it is common to specify the following:

a) **Delay time, (t_d):** The delay time is the time required for the response to reach half the final value the very first time.

b) **Rise time, (t_r):** The rise time is the time required for the response to rise from 10% to 90%, 5% to 95%, or 0% to 100% of its final value. For underdamped second- order systems, the 0% to 100% rise time is normally used. For overdamped systems, the 10% to 90% rise time is commonly used.

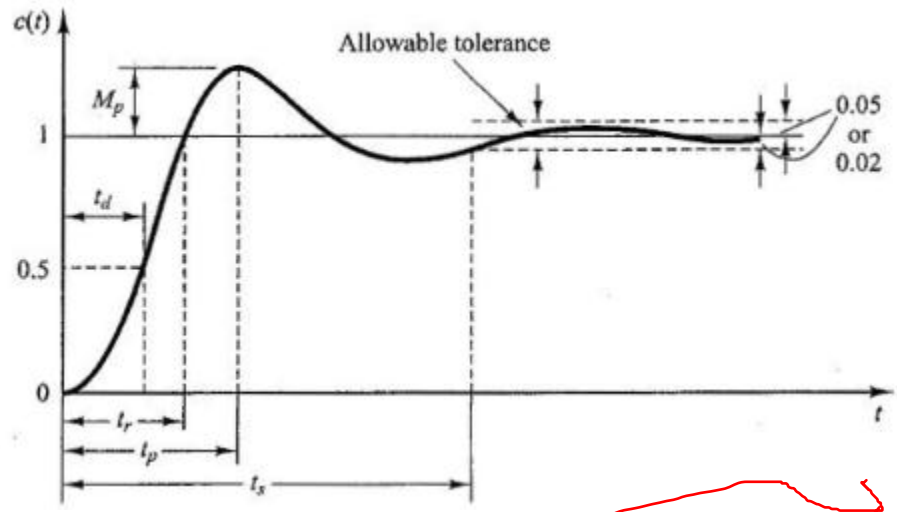
c) **Peak time, (t_p) :** The peak time is the time required for the response to reach the first peak of the overshoot.

d) **Maximum (percent) overshoot, (M_p):** The maximum overshoot is the maximum peak value of the response curve measured from unity. If the final steady-state value of the response differs from unity, then it is common to use the maximum percent overshoot. It is defined by

Maximum percent overshoot $= \frac{C(t_p)-C(\infty)}{C(\infty)}\ x\ 100\%$

The amount of the maximum (percent) overshoot directly indicates the relative stability of the system.

e) **Settling time,(t_s):** The settling time is the time required for the response curve to reach and stay within a range about the final value of size specified by absolute percentage of the final value (usually 2% or 5%). The settling time is related to the largest time constant of the control system. Which percentage error criterion to use may be determined from the objectives of the system design in question.



(A.5)

Second-Order Systems and Transient-Response Specifications

Rise timet_r: we obtain the rise time t_r by letting c (t_r) = 1

$$t_r = \frac{1}{\omega_d}\tan^{-1}\left(\frac{\omega_d}{-\sigma}\right) \qquad \omega_d = \omega_n\sqrt{1-\zeta^2} \qquad \sigma = \zeta\omega_n$$

Peak time t_p :

$$t_p = \frac{\pi}{\omega_d}$$

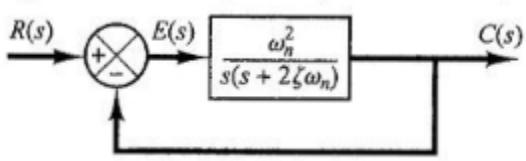
Settling time t_s :

$$t_s = 4T = \frac{4}{\sigma} = \frac{4}{\zeta\omega_n}\ \ (2\%\ Criterion)\ t_s = 3T = \frac{3}{\sigma} = \frac{3}{\zeta\omega_n}\ \ (5\%\ Criterion)$$

EXAMPLE

Consider the system shown in Figure, where ζ = 0.6 and ω_n = 5 rad/sec, Let us obtain the rise time t_r, peak time t_p maximum overshoot M_p, and settling time t_s when the system is subjected to a unit-step input.

From the given values of ζ and ω_n, we obtain $\omega_d = \omega_n\sqrt{1-\zeta^2}$ = 4 and $\sigma = \zeta\,\omega_n$ =3.



$t_r\ \frac{\pi-\beta}{\omega_d} = \frac{3.14-\beta}{4}$ $\quad \beta = \tan^{-1}\frac{\omega_d}{\sigma} = \tan^{-1}\frac{4}{3} = 0.93\ rad$ $\quad t_r = \frac{3.14-0.93}{4} = 0.55\ sec$

Peak time t_p: $\quad t_p = \frac{\pi}{\omega_d} = \frac{3.14}{4} = 0.785\ sec$

Maximum overshoot M_p: $\quad M_p = e^{-(\frac{\sigma}{\omega_d})\pi} = e^{-(\frac{3}{4})3.14} = 0.095$

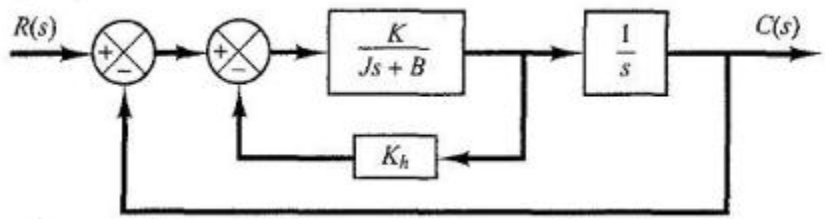
The maximum percent overshoot is thus 9.5 %.

Settling time t_s :

For the 2% criterion, the settling time is $t_s = \frac{4}{\sigma} = \frac{4}{3} = 1.33\ sec$ for 5% criterion, $t_s = \frac{3}{\sigma} = \frac{3}{3} = 1\ sec$

EXAMPLE 2

For the system shown in Figure, determine the values of gain K and velocity feedback constant K_h so that the maximum overshoot in the unit -step response is 0.2 and the peak time is 1 sec. With these values of K and K_h, obtain the rise time and settling time. Assume that J = 1kg-m^2 and B=1 N-m/rad/sec.



$$\frac{C(s)}{R(s)} = \frac{K}{Js^2 + (B + KK_h)s + K} \qquad \zeta = \frac{B + KK_h}{2\sqrt{KJ}} \qquad \omega_n = \sqrt{K/J}$$

The maximum overshoot M_p is given by $M_p = e^{-(\zeta/\sqrt{1-\zeta^2})\pi}$

This value must be 0.2. Thus,

$$e^{-(\zeta/\sqrt{1-\zeta^2})\pi} = 0.2 \qquad \frac{\zeta\pi}{\sqrt{1-\zeta^2}} = 1.61 \qquad \zeta = 0.456$$

The peak time t_p is specified as 1 sec

$$t_p = \frac{\pi}{\omega_d} = 1 \qquad \omega_d = 3.14 \qquad \omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}} = 3.53$$

Since the natural frequency ω_n is equal to $\sqrt{K/J}$, $\quad K = J\omega_n^2 = \omega_n^2 = 12.5$ N-m

$$K_h = \frac{2\sqrt{KJ}\zeta - B}{K} = \frac{2\sqrt{K}\zeta - 1}{K} = 0.178 \text{ sec}$$

$$t_r = \frac{\pi - \beta}{\omega_d} \qquad \beta = \tan^{-1}\frac{\omega_d}{\sigma} = \tan^{-1} 1.95 = 1.10 \qquad t_r = 0.65 \text{ sec}$$

Settling time t_s: For the 2% criterion, $\quad t_s = \dfrac{4}{\sigma} = 2.48$ sec

For the 5% criterion, $\quad t_s = \dfrac{3}{\sigma} = 1.86$ sec

HIGHER-ORDER SYSTEMS

Once the numerator and the denominator have been factored, $C(s)/R(s)$ can be written in the form

$$\frac{C(s)}{R(s)} = \frac{K(s + z_1)(s + z_2)\cdots(s + z_m)}{(s + p_1)(s + p_2)\cdots(s + p_n)}$$

Let us examine the response behavior of this system to a unit-step input. Consider first the case where the closed-loop poles are all real and distinct. For a unit-step input,

$$C(s) = \frac{a}{s} + \sum_{i=1}^{n} \frac{a_i}{s + p_i} \qquad \text{where } a_i \text{ is the residue of the pole at } s = -p_i$$

Dominant Closed-Loop Poles. The relative dominance of closed-loop poles is determined by the ratio of the real parts of the closed-loop poles, as well as by the relative magnitudes of the residues evaluated at the closed-loop poles. The magnitudes of the residues depend on both the closed-loop poles and zeros.

Stability of a System

The most important problem in linear control systems concerns stability. That is, under what conditions will a system become unstable? If it is unstable, how should we stabilize the system? It was stated that a control system is stable if and only if all closed loop poles lie in the left-half s plane.

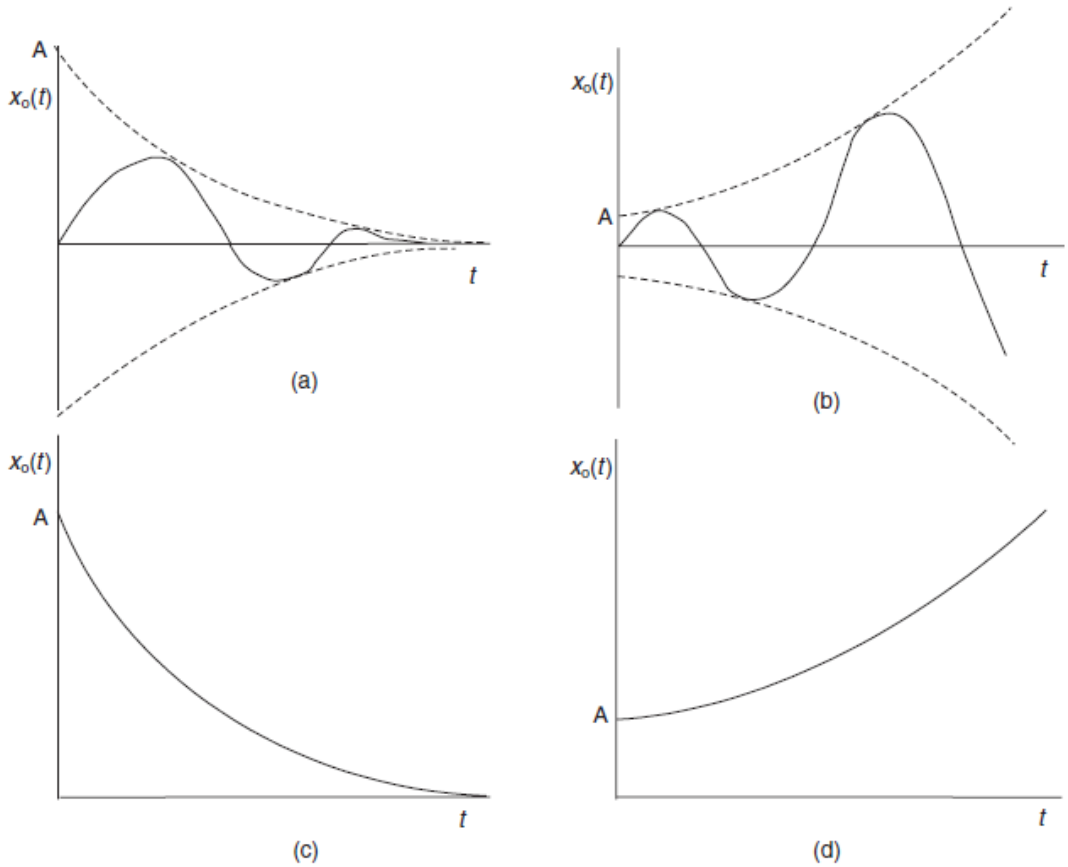


Figure 1: Stable and Unstable System

The Routh-Hurwitz Stability Criterion

The work of Routh and Hurwitz gives a method of indicating the presence and number of unstable roots, but not their value. Consider the characteristic equation.

$$a_n S^n + a_{n-1} S^{n-1} + \cdots + a_1 S + a_0 = 0$$

The Routh-Hurwitz stability criterion states

a) For there to be no roots with positive real parts then there is a necessary, but not sufficient, condition that all coefficients in the characteristic equation have the same sign and that none are zero.
If a) above is satisfied, then the necessary and sufficient condition for stability is either
b) All the Hurwitz determinants of the polynomial are positive, or alternatively
c) All coefficients of the first column of Routh's array have the same sign. The number of sign changes indicate the number of unstable roots

Routh table is arranged as follows

s^n	a_0	a_2	a_4	a_6	.	.	.
s^{n-1}	a_1	a_3	a_5	a_7	.	.	.
s^{n-2}	b_1	b_2	b_3	b_4	.	.	.
s^{n-3}	c_1	c_2	c_3	c_4	.	.	.
s^{n-4}	d_1	d_2	d_3	d_4	.	.	.
.	.	.					
.	.	.					
.	.	.					
s^2	e_1	e_2					
s^1	f_1						
s^0	g_1						

The coefficientsb_1, b_2, b_3, and so on, are evaluated as follows

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}, \quad b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}, \quad b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}, \ \ldots$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}, \quad c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}, \quad c_3 = \frac{b_1 a_7 - a_1 b_4}{b_1}, \ \ldots$$

$$d_1 = \frac{c_1 b_2 - b_1 c_2}{c_1}, \quad d_2 = \frac{c_1 b_3 - b_1 c_3}{c_1}, \ \ldots$$

Example: Check the stability of the system which has the following characteristic equation

$$S^4 + 2S^3 + S^2 + 4S + 2 = 0$$

S^4:	1	1	2
S^3:	2	4	
S^2:	-1	2	
S^1:	8		
S^0:	2		

In the array given, there are two sign changes in the column therefore there are two roots with positive real parts. Hence the system is unstable.

Root-locus analysis

The closed-loop transfer function for any feedback control system may be written in the factored form given in equation

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{K_c(s-z_{c1})(s-z_{c2})\ldots(s-c_{cm})}{(s-p_{c1})(s-p_{c2})\cdots(s-p_{cn})}$$

Where s =p_{c1},p_{c2},$\cdots$, p_{cn} are closed-loop poles and s =z_{c1},z_{c2},$\cdots$, z_{cm} are closed-loop zeros.

The position of the closed-loop poles in the s-plane determines the nature of the transient behavior of the system. The open-loop transfer function may be expressed as

$$G(s)H(s) = \frac{K(s-z_{o1})(s-z_{o2})\cdots(s-z_{om})}{(s-p_{o1})(s-p_{o2})\cdots(s-p_{on})}$$

Where z_{o1}, z_{o1}, $\cdots$, z_{om} are open-loop zeros and p_{o1}, p_{o2}, $\cdots$, p_{on} are open-loop poles.

Root-locus Plot

Angle and magnitude conditions: Consider the system's closed transfer function

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

The characteristic equation for this closed-loop system is obtained by setting the denominator of the right-hand side of the above equation. $1+G(s)H(s)$=0 or $G(s)H(s) = -1$, this equation can be split into two equations by equating the angle and magnitudes of both sides, respectively, to obtain the following

Angle Condition:

$$\angle G(s)H(s) = \pm 180°(2k+1) \qquad (k = 0, 1, 2, \ldots)$$

Magnitude Condition:

$$|G(s)H(s)| = 1$$

The values of s that fulfill both the angle and magnitude conditions are the roots of the characteristic equation.

<u>Root locus Construction Rules</u>

1. Starting points (K=0): The root loci start at the open-loop poles.
2. Termination points (K = ∞): The root loci terminate at the open-loop zero when they exist, otherwise at infinity
3. Symmetry of root loci: The root loci are symmetrical about the real axis. (Coefficients are real)
4. Root locus asymptotes: For large values of k the root loci are asymptotic to straight lines, with angles given by $\theta = \frac{(1+2K)}{(n-m)}$ X ±180 Where, k= 0, 1, … (n-m-1)

   n= no. of finite open-loop poles

   m= no. of finite open-loop zeros

5. Asymptote intersection: The asymptotes intersect the real axis at a point given by

$$\sigma_a = \frac{\sum open-loop\ poles - \sum open-loop\ zeros}{(n-m)}$$

6. Root locus location on real axis: A point on the real axis is part of the loci if the sum of the number of open-loop poles and zeros to the right of the point concerned is odd. (Why?)
7. Breakaway points: The point at which a locus breaks away from the real axis can be calculated using the following method

$$\left.\frac{dK}{ds}\right|_{s=\sigma_b} = 0$$ (Beakaway poins are multiput roots)

Where K has been made the subject of the characteristic equation i.e. K= …

8. Imaginary axis crossover: The location on the imaginary axis of the loci (marginal stability) can be calculated by replacing s by jw in the characteristic equation (since $\sigma = 0$ on the imaginary axis)
9. Angle of Departure and Arrival

Angle of departure from a complex pole = 180 **–** (sum of the angles of vectors to a complex pole in question from other poles) **+** (sum of the angles of vectors to a complex pole in question from zeros)

Angle of arrival at a complex zero = 180 **–** (sum of the angles of vectors to a complex zero in question from other zeros) **+** (sum of the angles of vectors to a complex zero in question from poles)

Example: Sketch the root locus of the following system

$$G(s)H(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+5)}$$

Solution:

$$G(s)H(s) = \frac{K(s+3)}{s(s+2)(s^2+4s+5)} = \frac{K(s+3)}{s(s+2)(s+2-j)(s+2+j)}$$

N=4, m=1, n-m=3

Real axis part of root locus: -2 $\leq s \leq 0$, $-\infty \leq s \leq -3$

Angle of asymptotes: $\theta_A = \frac{180(2q+1)}{3}$, $q = 0,1,2 \rightarrow \theta_A = 60°, 180°, 300° = \pm 60°, 180°$

Asymptote intersection: $\sigma_A = \frac{0+(-2)+(-2+j)+(-2-j)-(-3)}{3} = -1$

Breakaway point: $1 + \frac{K(s+3)}{s(s+2)(s^2+4s+5)} = 0$ $K = \frac{-(s^4+6s^3+13s^2+10s)}{s+3}$,

$$\frac{dK}{ds} = 0,\ \sigma_B = -0.7509, -3.5424$$

Angle of Departure:

$$\theta_d = -[\,180° + 90° + \theta_{p3} + \theta_{p4} - \theta_{z1}]$$

$$= -[270° + \tan^{-1}\left(\frac{1-0}{-2-0}\right) + \tan^{-1}\left(\frac{1-0}{-1-(-2)}\right) - \tan^{-1}\left(\frac{1-0}{-2-(-3)}\right)]$$

$\theta_d = -468.4349° => \ \theta_d = -108.4349°$

Imaginary Crossover:

F(s)= $s^4 + 6s^3 + 13s^2 + (10 + K)s \ + 3K$

F(s) = $(jw)^4 + 6(jw)^3 + 13(jw)^2 + (10 + K)(jw) \ + 3K$=0

F(s) = $w^4 - \ 6j\ w^3 - \ 13w^2 + j(10 + K)w \ + 3K$=0

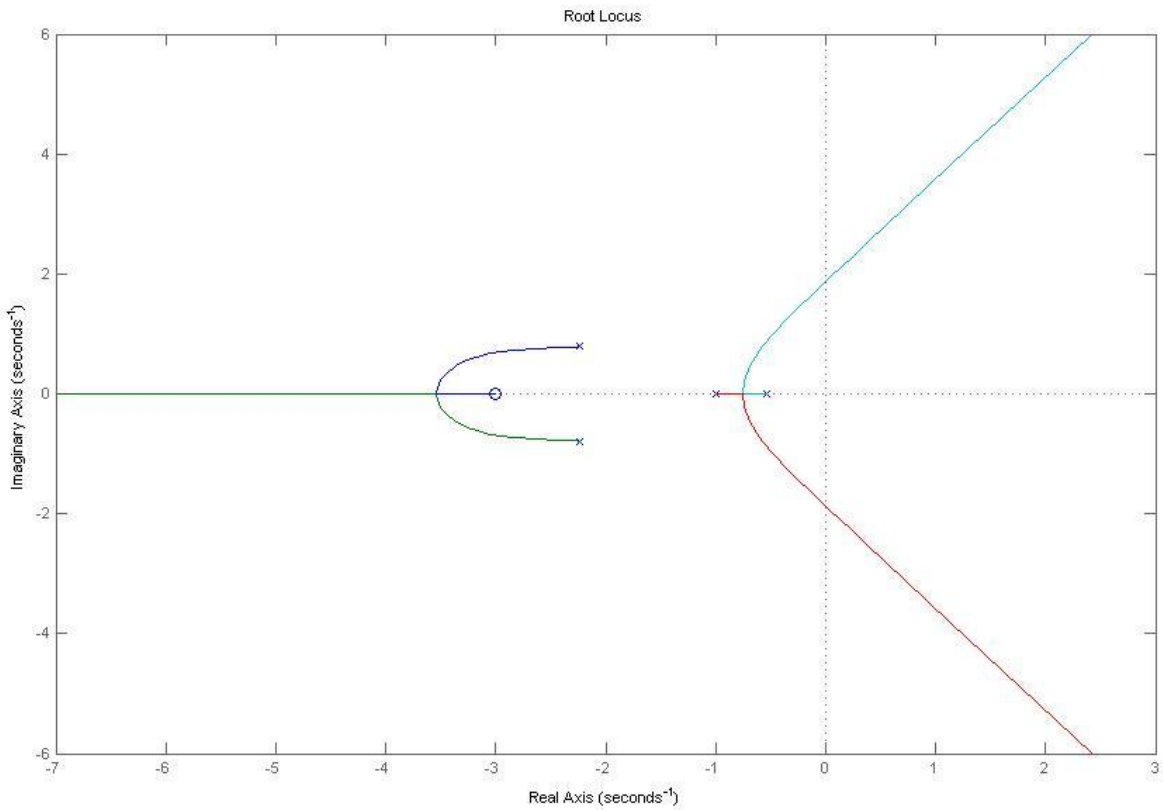
Equating imaginary part

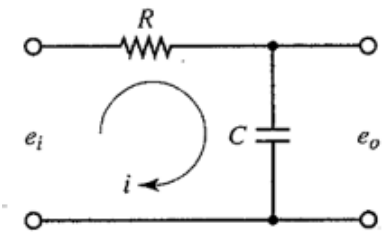
$-6w^3$ +(10+K)w =0

Equating real parts:

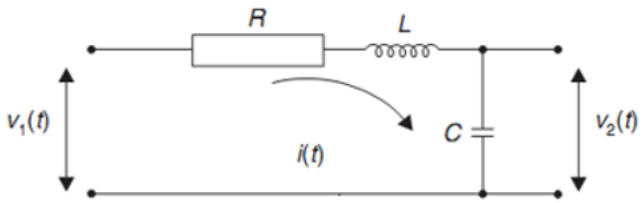
$w^4 \ - 13w^2$ +3K=0

After Solving the above equations, w=1.88 and k=11.1





1. FindTime constant



2. Assuming this system is underdamped.
Determine the following if input is a unit-step
- natural frequency , damping ratio
- If input is a unit step, find
+ peak amplitude
+ rise time (100%)
+ settling time(2% criteria)

Terminology

- transient / steady state respone
- first order / second order / higher order
- damped / undamped natural frequency , damping ratio
- overdamped / critical / underdamped system
- delay time / rising time / peak time / settling time ...peak amplitude maximum over shoot
- stability (BIBO) poles / zeros
- Routh-Hurwitz
- Root Locus